

Introduction to Forest Ecology

What is Forest Ecology?

Forest ecology is the study of forests as complex ecosystems. It examines the interactions between trees, plants, animals, and their physical environment.

Key Concepts

- **Biodiversity:** Forests harbour the majority of terrestrial species on Earth
 - **Nutrient cycling:** Decomposition recycles organic matter back into the soil
 - **Succession:** Forest communities change over time following disturbance
 - **Disturbance regimes:** Fire, wind, and insect outbreaks shape forest structure
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Climate Change Impacts on Forests

Rising Temperatures

Average global temperatures have increased by approximately 1.1 degrees Celsius since pre-industrial times. Forests in boreal and temperate zones are particularly sensitive.

Effects on Biodiversity

- Shifts in species ranges toward higher latitudes and elevations
- Increased frequency of drought stress and tree mortality
- Altered phenology, including earlier leaf-out and longer growing seasons
- Decline of climate-sensitive species such as mountain pine beetle host trees

Carbon Storage

Forests act as carbon sinks, absorbing roughly 2.6 billion tonnes of carbon dioxide annually. Climate change threatens this function through increased wildfire frequency and drought-induced dieback.

Forest Management Strategies

Sustainable Harvesting

- Selective logging to maintain canopy cover and wildlife habitat
- Rotation periods matched to species growth rates
- Retention of legacy trees for structural diversity

Adaptive Management

- Monitoring forest health indicators over time

- Adjusting harvest plans based on climate projections
- Integrating Indigenous ecological knowledge into stewardship practices

Restoration Ecology

- Reforestation of degraded lands using native species
- Rehabilitation of riparian buffers to protect water quality
- Invasive species control to restore natural community composition